Supplement Materials

Table S1. Top 5 features with the highest correlation with *Diagnosis*.

Feature	Correlation
FastingBloodSugar	0.4520
HbA1c	0.4311
FrequentUrination	0.1515
Hypertension	0.1319
ExcessiveThirst	0.0735

Table S2. Distribution of *Diagnosis* classes.

Diagnosis	**Count**	**Percentage**
No Diabetes (0)	1127	59.98
Diabetes (1)	752	40.02

Table S3. Comparison of ROC-AUC for best cases and average value during hyperparameter tuning for all models. This evaluates model stability. Models are: Random Forest (RF), Gradient Boosting (GB), Logistic Regression (LR), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN).

Model	CV ROC-AUC	ROC-AUC	Decrease Percentage
RF	0.9549	0.9589	0.42
GB	0.9601	0.9636	2.89
LR	0.9029	0.9116	0.95
SVM	0.8978	0.9245	0.36
KNN	0.8615	0.8988	4.15

Table S4. Detailed coefficient values of Logistic Regression (features scored 0 are not listed here).

Feature	Coefficient (Signed Value)	Coefficient (Absolute Value)
FastingBloodSugar	1.5032	1.5032
HbA1c	1.4084	1.4084
FrequentUrination	1.0396	1.0396
Hypertension	1.0231	1.0231
UnexplainedWeightLoss	0.5749	0.5749
ExcessiveThirst	0.4307	0.4307
SocioeconomicStatus_0	-0.1809	0.1809
SystolicBP	-0.0818	0.0818
CholesterolTriglycerides	0.0645	0.0645
QualityOfLifeScore	0.0302	0.0302
BMI	0.0276	0.0276
BUNLevels	0.0251	0.0251

SleepQuality	-0.0085	0.0085
DietQuality	-0.0010	0.0010

Table S5. Detailed feature importance values of Random Forest.

Feature	Importance
FastingBloodSugar	0.3040
HbA1c	0.2922
FrequentUrination	0.0237
HealthLiteracy	0.0187
SystolicBP	0.0184
BMI	0.0183
Hypertension	0.0182
DietQuality	0.0178
QualityOfLifeScore	0.0174
CholesterolTotal	0.0172
DiastolicBP	0.0171
CholesterolHDL	0.0167
SerumCreatinine	0.0166
PhysicalActivity	0.0163
SleepQuality	0.0162
FatigueLevels	0.016
CholesterolLDL	0.0157
MedicationAdherence	0.0156
CholesterolTriglycerides	0.0153
BUNLevels	0.0152
AlcoholConsumption	0.0148
MedicalCheckupsFrequency	0.0147
Age	0.0142
ExcessiveThirst	0.0056
UnexplainedWeightLoss	0.0046
Smoking	0.0027
EducationLevel_2	0.0023
Gender	0.0023
Ethnicity_0	0.0022
Statins	0.0022
FamilyHistoryDiabetes	0.0019
EducationLevel_3	0.0019
SocioeconomicStatus_0	0.0019
AntidiabeticMedications	0.0018
SocioeconomicStatus_2	0.0018
SocioeconomicStatus_1	0.0018
EducationLevel_1	0.0017
AntihypertensiveMedications	0.0016

Ethnicity_1	0.0015
BlurredVision	0.0014
SlowHealingSores	0.0013
GestationalDiabetes	0.0013
WaterQuality	0.0012
Ethnicity_2	0.0010
EducationLevel_0	0.0010
TinglingHandsFeet	0.0010
Ethnicity_3	0.0009
PreviousPreDiabetes	0.0009
OccupationalExposureChemicals	0.0008
PolycysticOvarySyndrome	0.0007
HeavyMetalsExposure	0.0005

Table S6. Detailed feature importance values of Gradient Boosting (features scored 0 are not listed here).

Feature	**Importance**
FastingBloodSugar	0.4169
HbA1c	0.3883
FrequentUrination	0.0414
Hypertension	0.0346
UnexplainedWeightLoss	0.0140
ExcessiveThirst	0.0120
FatigueLevels	0.0078
SleepQuality	0.0077
HealthLiteracy	0.0066
SystolicBP	0.0065
SerumCreatinine	0.0055
DietQuality	0.0054
CholesterolTotal	0.0050
Age	0.0046
BMI	0.0046
BUNLevels	0.0044
DiastolicBP	0.0041
PhysicalActivity	0.0038
AlcoholConsumption	0.0036
QualityOfLifeScore	0.0034
MedicationAdherence	0.0034
MedicalCheckupsFrequency	0.0031
CholesterolTriglycerides	0.0027
CholesterolHDL	0.0022
Ethnicity_3	0.0016

Smoking	0.0014
CholesterolLDL	0.0011
SocioeconomicStatus_0	0.0008
Statins	0.0007
EducationLevel_1	0.0005
SocioeconomicStatus_2	0.0005
Ethnicity_0	0.0004
BlurredVision	0.0004
EducationLevel_2	0.0004
Ethnicity_1	0.0003
FamilyHistoryDiabetes	0.0002
HeavyMetalsExposure	0.0002
EducationLevel_3	0.0001
EducationLevel_0	0.0001
OccupationalExposureChemicals	0.0001

Table S7. Detailed SHAP values (mean absolute values) of Random Forest.

Feature	**SHAP Value**
HbA1c	0.1685
FastingBloodSugar	0.1666
FrequentUrination	0.0301
Hypertension	0.0218
SystolicBP	0.0084
ExcessiveThirst	0.0067
DiastolicBP	0.0061
DietQuality	0.0059
QualityOfLifeScore	0.0056
BMI	0.0050
UnexplainedWeightLoss	0.0043
CholesterolTotal	0.0042
CholesterolHDL	0.0032
MedicationAdherence	0.0032
HealthLiteracy	0.0031
BUNLevels	0.0031
AlcoholConsumption	0.0031
CholesterolTriglycerides	0.0031
PhysicalActivity	0.0029
SleepQuality	0.0029
Age	0.0027
Smoking	0.0026
FatigueLevels	0.0025
SerumCreatinine	0.0025
CholesterolLDL	0.0023

MedicalCheckupsFrequency	0.0020
SocioeconomicStatus_0	0.0015
FamilyHistoryDiabetes	0.0013
BlurredVision	0.0012
SocioeconomicStatus_2	0.0011
EducationLevel_2	0.0011
Ethnicity_0	0.0009
GestationalDiabetes	0.0009
EducationLevel_3	0.0008
WaterQuality	0.0007
PolycysticOvarySyndrome	0.0007
Statins	0.0007
SocioeconomicStatus_1	0.0006
EducationLevel_1	0.0006
Ethnicity_1	0.0006
Gender	0.0006
TinglingHandsFeet	0.0005
AntidiabeticMedications	0.0005
Ethnicity_3	0.0005
Ethnicity_2	0.0004
SlowHealingSores	0.0004
AntihypertensiveMedications	0.0004
OccupationalExposureChemicals	0.0003
EducationLevel_0	0.0003
PreviousPreDiabetes	0.0003
HeavyMetalsExposure	0.0002

Table S8. Detailed SHAP values (mean absolute values) of Gradient Boosting (features scored 0 are not listed here).

Feature	**SHAP Value**
FastingBloodSugar	1.9073
HbA1c	1.8705
FrequentUrination	0.4878
Hypertension	0.3563
ExcessiveThirst	0.2847
UnexplainedWeightLoss	0.1869
DietQuality	0.0777
BMI	0.0776
CholesterolTotal	0.0773
Age	0.0559
BUNLevels	0.0546
SystolicBP	0.0533

CholesterolTriglycerides	0.0528
DiastolicBP	0.0469
QualityOfLifeScore	0.0445
SleepQuality	0.0437
HealthLiteracy	0.0401
FatigueLevels	0.0387
MedicalCheckupsFrequency	0.0384
Smoking	0.0376
SocioeconomicStatus_0	0.0332
SocioeconomicStatus_2	0.0301
MedicationAdherence	0.0249
AlcoholConsumption	0.0245
SerumCreatinine	0.0241
PhysicalActivity	0.0238
BlurredVision	0.0192
EducationLevel_2	0.0148
CholesterolHDL	0.0132
CholesterolLDL	0.0105
Ethnicity_0	0.0104
EducationLevel_3	0.0081
Ethnicity_3	0.0063
EducationLevel_1	0.0062
Statins	0.0052
FamilyHistoryDiabetes	0.0047
OccupationalExposureChemicals	0.0029
Ethnicity_1	0.0027
HeavyMetalsExposure	0.0015
EducationLevel_0	0.0014

Table S9. Detailed permutation importance values for each of the five models and mean scores. Negative values are indicated in italics. Models are: Random Forest (RF), Gradient Boosting (GB), Logistic Regression (LR), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN).

Feature	RF	SVM	LR	GB	KNN	Importance
HbA1c	0.2251	0.2052	0.2021	0.2211	0.1972	0.2101
FastingBloodSugar	0.2253	0.1951	0.1915	0.2207	0.1928	0.2051
FrequentUrination	0.0157	0.0224	0.0186	0.0178	0.0067	0.0162
Hypertension	0.0137	0.0144	0.0129	0.0124	0.0069	0.0121
ExcessiveThirst	0.0026	0.0106	0.0075	0.007	0.0016	0.0059
CholesterolTotal	0.0012	*0.0007*	0	0.0014	0.0166	0.0037
CholesterolHDL	0.0027	0.0031	0	0.0002	0.0096	0.0031
CholesterolLDL	0.0011	0.0044	0	0.0001	0.0089	0.0029
HealthLiteracy	*0.0008*	*0.0016*	0	0.0003	0.0164	0.0029

MedicalCheckupsFrequency	0.0014	0.0011	0	*0.0004*	0.0122	0.0029
DietQuality	0.0009	0.0032	0	0.0013	0.0084	0.0028
QualityOfLifeScore	0.0014	0.0018	0	0.0014	0.0079	0.0025
FamilyHistoryDiabetes	0.0007	0	0	0	0.0118	0.0025
BMI	*0.0005*	*0.0002*	*0.0001*	0.0001	0.0129	0.0024
Gender	*0.0001*	0.0008	0	0	0.0114	0.0024
MedicationAdherence	*0.0004*	*0.0003*	0	0.0003	0.0111	0.0021
DiastolicBP	0.0012	0	0	0	0.0087	0.002
AlcoholConsumption	0.0002	*0.0003*	0	*0.0004*	0.01	0.0019
PhysicalActivity	*0.0003*	*0.0015*	0	0.0001	0.0101	0.0017
CholesterolTriglycerides	0.0009	0.0009	0.0006	0.0012	0.0048	0.0017
WaterQuality	0.0003	0.0022	0	0	0.0055	0.0016
SocioeconomicStatus_1	0	*0.0001*	0	0	0.0078	0.0015
SystolicBP	*0.0005*	0.0032	0.0004	*0.0001*	0.0038	0.0014
SerumCreatinine	0.0006	0.0013	0	0.0016	0.0031	0.0013
Statins	0.0003	0.0011	0	0.0001	0.0049	0.0013
AntihypertensiveMedications	0.0002	0.0006	0	0	0.0055	0.0013
Age	*0.0002*	*0.0003*	0	*0.0013*	0.0077	0.0012
SocioeconomicStatus_2	*0.0004*	*0.0005*	0	0	0.0058	0.001
BlurredVision	0.0001	0.0023	0	0.0007	0.0013	0.0009
SocioeconomicStatus_0	0.0001	*0.0005*	0.0001	*0.0003*	0.0048	0.0008
Smoking	0.0002	0.0001	0	0.0004	0.0034	0.0008
EducationLevel_0	*0.0001*	0.0007	0	0	0.0034	0.0008
UnexplainedWeightLoss	0.0002	*0.0005*	0	0.0043	0	0.0008
AntidiabeticMedications	*0.0003*	*0.0003*	0	0	0.0039	0.0007
EducationLevel_3	0.0001	0	0	0.0002	0.0019	0.0004
BUNLevels	*0.0002*	*0.0001*	0.0002	0.0001	0.0021	0.0004
SleepQuality	*0.0002*	*0.0032*	0	0.0004	0.0049	0.0004
GestationalDiabetes	0.0003	0.001	0	0	0.0005	0.0004
HeavyMetalsExposure	0.0001	0.0002	0	0	0.0013	0.0003
Ethnicity_0	*0.0001*	*0.0011*	0	0.0001	0.0025	0.0003
TinglingHandsFeet	*0.0001*	0.0001	0	0	0.0009	0.0002
SlowHealingSores	0.0001	0.0005	0	0	*0.0002*	0.0001
PolycysticOvarySyndrome	*0.0002*	0.0003	0	0	*0.0005*	*0.0001*
FatigueLevels	*0.0008*	*0.0031*	0	0.0001	0.0033	*0.0001*
Ethnicity_2	0.0003	*0.0001*	0	0	0.0012	*0.0002*
OccupationalExposureChemicals	*0.0002*	*0.0012*	0	0	0.0007	*0.0004*
EducationLevel_2	0	0	0	0	*0.0023*	*0.0005*
PreviousPreDiabetes	0	0.0003	0	0	*0.003*	*0.0005*
EducationLevel_1	0.0002	*0.0003*	0	0.0001	*0.0031*	*0.0006*
Ethnicity_3	*0.0003*	*0.0008*	0	*0.0003*	*0.0033*	*0.0009*
Ethnicity_1	*0.0001*	*0.0013*	0	0	*0.0048*	*0.0012*